

PHY F376: A study of hybrid classical-quantum algorithms

Week 6 (25/9/2026) - A Chemistry-Informed Ansatz and a Completed Hardware Comparison

Khushi Pandey (2023B5A70951G)

Overview of This Week's Work

This week addressed the detailed feedback on last week's H₂O CAS(4e,4o) results directly: replacing the hand-rolled ansatz with a proper chemistry-informed circuit, fixing several real bugs in the workflow, and finally completing a matched 10,000-shot run on both the noiseless simulator and real IBM hardware.

A Chemistry-Informed, Reproducible Ansatz

Since no separate integrals or FCIDUMP file exist for this system, CCSD-amplitude seeding was not possible, but the state-preparation circuit could still be built from `ffsim`'s actual LUCJ architecture rather than a hand-designed one, with parameters optimized directly against the Hamiltonian in place of amplitude seeding. Rather than picking a circuit depth by hand, the code now sweeps `n_reps` $\in \{1, 2, 3\}$ and selects the best option by an explicit rule, so the choice is reproducible from the code alone rather than resting on judgement made outside it. This sweep selected $n_{reps} = 3$ (error 7.15 mHa, two-qubit depth 27), a large improvement over the previous hand-rolled ansatz's depth of 105-124 at worse accuracy.

The Completed Hardware Run

With both simulator and hardware run at 10,000 shots each, the two are now directly comparable. The hardware circuit transpiled to a two-qubit depth of 57 (against the real backend, deeper than the synthetic stand-in used during development), and the job's ID, physical qubit layout, raw counts, and backend calibration snapshot were saved alongside the run. Correcting an earlier bug, the fraction of shots landing in the correct particle-number sector and the fraction of distinct bitstrings that are valid are now reported separately: 52.74% and 14.11% respectively for hardware, against 100% for both on the simulator. SQD converged to about 2.7 mHa on the simulator, and to a noticeably noisier 10-18 mHa range on hardware, consistent with the noise level implied by those retained fractions.

Shot Dependence and Batch-Size Convergence

Repeated random subsampling of the existing 10,000-shot data (no new QPU time spent) showed the simulator's mean SQD error dropping fairly steadily from about 19.9 mHa at 500 shots to 3.8 mHa at 10,000, while the hardware curve was noisier and non-monotonic (10.2 to 14.9 to 4.6 mHa), which is expected given real noise rather than a sign of a bug. Sweeping `samples_per_batch` from 5 to 36 gave a clean, monotonic convergence on both datasets; notably, the hardware sweep actually reached a lower final error (0.61 mHa, using 35 of the 36 valid configurations) than the simulator (which plateaued at 2.67 mHa, having only sampled 12 unique valid configurations), since hardware noise, despite lowering the valid-shot fraction, also spread sampling across a wider variety of the correct-sector states.

Interpreting the Hardware Result

The gap between the 52.74% shot-weighted valid fraction and the 14.11% distinct-bitstring fraction is itself informative: it shows that while noise causes the hardware to sample a much wider variety of bitstrings than the simulator, most of the actual shot probability mass still concentrates on a smaller set of physically valid configurations. This is consistent with the circuit genuinely encoding correct chemical information, with noise acting more as a broadening effect than a source of systematic bias.

Concluding Remarks

The single-circuit SQD benchmark on H₂O CAS(4e,4o) is now complete, reproducible, and directly comparable between simulator and hardware, with the earlier reporting bugs resolved and the ansatz choice justified by code rather than by hand. With this in place, the next step is deciding whether to move to SKQD or to a larger active space.